

Introduction to Printing Technology

Prepress • Press • Postpress • Screen Printing Operations • Laboratory Safety & Discipline

POLY PMNA

Course Code	1101
Programme	Printing Technology
Contact Hours	90
L:T:P:S	3:0:3:0
Self-Learning (SS)	6 hrs/week
Credits	4.5
Prerequisites	NIL
Bloom Level	Apply

Practicum Focus: Prepress design/typesetting, press ink-transfer, postpress finishing, and single/multi-colour screen printing with registration safety.

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 1101 — Introduction to Printing Technology**. Use the official SITTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

Official SITTTR Revision 2026 syllabus reference. This page is a study handbook, not a substitute for the latest official notification or examination instruction.

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

OFFICIAL SOURCE DECLARATION & TRANSPARENCY

Verbatim SITTTR Syllabus Data & Disclosed Discrepancies

This handbook is compiled strictly from the official SITTTR Kerala Diploma Revision 2026 Course Description page for Subject Code 1101. To ensure 100% academic honesty, all verified inconsistencies in the official source are explicitly declared below.

1. Conflicting Assessment Scheme Values in Source

The official syllabus document contains two conflicting evaluation tables:

- **Top "COURSE DESCRIPTION" Table:** States CIE Marks = 40, ESE Marks = 100 (Total 140).
- **Bottom "Teaching and Assessment Scheme" Table:** Itemises Theory: CIE 20 + ESE 60 = 80, Practical: CIE 20 + ESE 40 = 60 (Grand Total = 140).

Disclosed Status: Both tables have been reproduced verbatim in this handbook. Students and faculty should consult SITTTR board circulars regarding the internal/external mark distribution for evaluation.

2. Verified Typographical Value in CO-PO Matrix

In the official CO-PO Articulation Matrix, the **Correlation Level** row for column P03 is printed exactly as 1,5 (using a comma).

Disclosed Status: This value has been preserved as 1,5 exactly as printed in the official source data, flagged as a probable typographical error for 1.5.

⚠ **Missing Source Data Disclosure:** The supplied official syllabus PDF for 1101 contains only the front course-description page. It does *NOT* contain module titles, module-wise topic lists, module hour allocations, official practical job lists, prescribed textbooks, or references. As required by quality rules, these sections are explicitly marked **"Not**

specified in the supplied official syllabus PDF — pending module/practical/reference pages" and have not been fabricated.

COURSE DASHBOARD

Course Structure and Assessment Scheme

Subject 1101 introduces printing technology, equipment, workflow stages, and screen printing practical operations in Semester I of the Diploma in Printing Technology programme.

90

Contact hours

4.5

Credits

3:0:3:0

L : T : P : S scheme

Practicum

Course type

Teaching and Assessment Scheme (Verbatim SITTTR Table)

Teaching Scheme / Week	Self-Learning (SS) / Week	Total Hours / Semester	Credits (C)	Examination Scheme (Theory)	Examination Scheme (Practical)	Grand Total
L = 3, T = 0, P = 3, S = 0	6 Hours	90 Hours	4.5	CIE: 20 ESE: 60 (Total: 80)	CIE: 20 ESE: 40 (Total: 60)	140 Marks

Footnote (verbatim): L: Lecture (1 hr = 1 credit), T: Tutorial (1 hr = 1 credit), P: Practical (1 hr = 0.5 credit), S: Project (1 hr = 0.5 credit), SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination.

Course Prerequisites

Topic	Course Code	Course Title	Semester
NIL	(blank)	(blank)	(blank)

OBJECTIVES & COURSE OUTCOMES

Curriculum Goals and Competency Mapping

Course Objectives (Verbatim)

1. Understand the role and importance of printing in daily life
2. Demonstrate basic printing workflow activities (prepress, press, postpress)
3. Identify basic printing processes, machines, and materials
4. Identify fundamental printing and finishing equipment and its components
5. Follow safe and disciplined laboratory practices

6. Perform basic screen printing operations

Course Outcomes (Verbatim)

- **CO1:** Explain the role and evolution of printing in society and industry — *Bloom's Level: Understand*
- **CO2:** Demonstrate basic prepress, press and postpress workflow activities — *Bloom's Level: Apply*
- **CO3:** Identify printing processes and Practice laboratory safety, discipline, and teamwork — *Bloom's Level: Apply*
- **CO4:** Perform single and multi-colour screen printing with proper registration and safety practices — *Bloom's Level: Apply*

CO-PO Mapping: Course Articulation Matrix (Verbatim)

CO-PO Articulation Matrix											
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	1	-	-	-	1	-	-	-	1	1
CO2	2	2	1	1	1	-	-	-	1	1	-
CO3	2	1	-	-	1	1	1	2	2	-	-
CO4	2	2	2	1	1	-	-	1	1	-	-
Total	8	6	3	2	3	2	1	3	4	2	1
Correlation Level	2	1.5	1,5*	1	1	1	1	1.5	1.33	1	1

Legend: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-".

*Note: Value 1,5 for PO3 is reproduced exactly as printed in the official SITTTT syllabus document (probable typo for 1.5).

COVERAGE MATRIX

Internal Syllabus Coverage Status

Confirmed Syllabus Components vs Pending Content

PDF Source Section	Component	Official Topic / Data	Hours	Target CO	Bloom Level	Handbook Section Status
Course Info	Subject Identity	1101 — Introduction to Printing Technology	90	CO1-CO4	Understand / Apply	Fully Populated (Cover & Dashboard)
Introduction	Course Scope	Role of printing, prepress/press/postpress, lab safety	90	CO1-CO4	Understand / Apply	Fully Populated (Section 1-4)
Objectives	Objectives 1-6	Role in daily life, workflow, processes, safety, screen printing	90	CO1-CO4	Understand / Apply	Fully Populated (Section 4)
Outcomes	CO1-CO4	Role/evolution, workflow, processes/safety, screen printing	90	CO1-CO4	Understand / Apply	Fully Populated (Section 4)
Assessment	Scheme Tables	CIE 40 / ESE 100 vs Theory 80 / Practical 60	90	CO1-CO4	Apply	Fully Populated & Disclosed
Module Pages	Module 1 Breakdown	Pending — awaiting module-content pages of official syllabus	Pending	CO1	Understand	Pending Notice Inserted (Module 1)
Module Pages	Module 2 Breakdown	Pending — awaiting module-content pages of official syllabus	Pending	CO2	Apply	Pending Notice Inserted (Module 2)
Module Pages	Module 3 Breakdown	Pending — awaiting module-content pages of official syllabus	Pending	CO3	Apply	Pending Notice Inserted (Module 3)
Module Pages	Module 4 Breakdown	Pending — awaiting module-content pages of official syllabus	Pending	CO4	Apply	Pending Notice Inserted (Module 4)

PDF Source Section	Component	Official Topic / Data	Hours	Target CO	Bloom Level	Handbook Section Status
Practical Pages	Experiment Job List	Pending — awaiting official screen-printing practical job list	45 (P)	CO4	Apply	Pending Notice Inserted (Section 8 & 11)

LEARNING ROADMAP

Provisional Workflow Progression Overview

1. Role of Printing in Society (CO1) →

2. Prepress, Press & Postpress (CO2) →

3. Printing Processes & Safety (CO3) →

4. Screen Printing Practice & Registration (CO4) →

5. Visual Printing Lab →

6. Practice Questions

MODULE CHAPTERS

Syllabus Content & Fundamental Orientation

CO1 Target

Bloom: Understand

Theory / Orientation

1.1 Importance of Printing in Daily Life and Communication–

Malayalam explanation: പ്രിന്റിംഗ് (Printing) പരസ്യം (Advertising), ലജിസ്റ്റിക്സ് (Logistics), മീഡിയ (Media) എന്നിവയെ സംബന്ധിച്ചതാണ്. Technical terms: *Packaging, Advertising, Logistics, Media.*

1.2 Evolution of Printing Technologies–

[illegible]

Basic Printing Workflow Activities (Prepress, Press & Postpress)

Demonstrating the three fundamental phases of print manufacturing from initial design to finished product.

CO2 Target

Bloom: Apply

Workflow Sequence

Not specified in the supplied official syllabus PDF — pending module/practical/reference pages: Detailed topic list and hour allocations for Module II are pending syllabus page release. Content below reflects confirmed CO2 and Objective 2.

2.1 Prepress Workflow (Design, Layout & Plate Making)–

Prepress encompasses all activities prior to actual printing on press. Key steps include graphic design, image capture, typesetting, colour separation (CMYK), page imposition, computer-to-plate (CTP) imaging or stencil preparation for screen printing.

Malayalam explanation: പ്രിന്റിംഗ് പ്രക്രിയയുടെ ആദ്യ ഘട്ടമാണ് പ്രീപ്രസ്സ്. ഗ്രാഫിക് ഡിസൈൻ, ഇമേജ് ക്യാപ്ചർ, ടൈപ്പെറ്റിംഗ്, കോളർ സെപ്പറേഷൻ (CMYK), പേജ് ഇംപോസിഷൻ, കമ്പ്യൂട്ടർ-ടു-പ്ലേറ്റ് (CTP) ഇമേജിംഗ് / സ്റ്റൈൻ പ്രിപ്പറേഷൻ എന്നിവയെല്ലാം പ്രീപ്രസ്സ് പ്രക്രിയയുടെ ഭാഗമാണ്.

2.2 Press Operations (Ink Transfer & Printing Systems)–

The Press stage involves transferring ink from an image carrier (plate, cylinder, screen) onto a substrate (paper, plastic, fabric, metal). It includes sheet feeding, ink metering, damping (in offset), cylinder impression, and register control.

Malayalam explanation: പ്രിന്റിംഗ് പ്രക്രിയയുടെ രണ്ടാം ഘട്ടമാണ് പ്രസ്സ് ഘട്ടം. ഇംക് ട്രാൻസ്ഫർ, ഇംക് മീറ്ററിംഗ്, ഡാമ്പിംഗ് (ഓഫ്സെറ്റിൽ), സിലിണ്ടർ ഇംപ്രഷൻ, റിജിസ്റ്റർ കൺട്രോൾ എന്നിവയെല്ലാം പ്രസ്സ് ഘട്ടത്തിൽ ഉൾപ്പെടുന്നു.

2.3 Postpress Operations (Finishing & Binding)–

Postpress includes all finishing operations performed after printing: guillotine cutting, folding, collating, stitching/binding, trimming, lamination, die-cutting, and foil stamping to produce final books, boxes, or cards.

Malayalam explanation: പരസ്പരം വിരുദ്ധമായ Postpress (Finishing). പരസ്പരം, പരസ്പരം, പരസ്പരം/പരസ്പരം, പരസ്പരം, പരസ്പരം പരസ്പരം പരസ്പരം.

Printing Processes, Machines, Materials & Laboratory Safety

Identification of major printing processes and strict enforcement of laboratory safety, discipline, and teamwork.

CO3 Target

Bloom: Apply

Processes & Safety

Not specified in the supplied official syllabus PDF — pending module/practical/reference pages: Topic details for Module III are pending syllabus expansion. Content below reflects confirmed CO3, Objectives 3, 4 & 5.

3.1 Major Printing Processes & Image Carriers–

Major Printing Processes Comparison			
Process	Image Carrier Surface	Ink Transfer Principle	Typical Products
Relief (Flexo / Letterpress)	Raised image areas	Ink applied to raised surface and pressed onto substrate	Labels, corrugated boxes, packaging
Planographic (Offset)	Flat surface (Chemical balance)	Ink accepts oil-receptive image; water repels non-image	Textbooks, newspapers, magazines
Intaglio (Gravure)	Sunken / Recessed cells	Ink filled in cells; excess wiped by doctor blade	High-volume packaging, currency
Stencil (Screen Printing)	Porouse mesh screen	Ink squeezed through open mesh areas with a squeegee	Textiles, T-shirts, signboards, circuit boards

Malayalam explanation: രണ്ട് തരത്തിൽ പ്രിന്റിംഗ് പ്രക്രിയകൾ ഉണ്ട്: Relief (രേഖാപ്രിന്റിംഗ്), Planographic/Offset (പ്ലാനോഗ്രാഫിക്/ഓഫ്സെറ്റ്), Intaglio/Gravure (ഇന്റാല്യോ/ഗ്രാവ്യൂർ പ്രിന്റിംഗ്), Stencil/Screen Printing (സ്റ്റൈൻസിൽ/സ്ക്രീൻ പ്രിന്റിംഗ്).

3.2 Laboratory Safety Rules, Discipline & Teamwork–

Mandatory Laboratory Safety Rules:

- Wear protective lab aprons and closed leather footwear at all times.
- Keep long hair tied back; remove loose jewellery, ties, and hanging clothing near rotating press rollers.
- Ensure proper room ventilation when handling solvents, wash-up chemicals, and photo emulsions.
- Know the exact location of Emergency Stop buttons, eyewash stations, and fire extinguishers.
- Maintain discipline, cleanliness, and active teamwork during practical lab exercises.

Malayalam explanation: തിരുവനന്തപുരം നഗരത്തിൽ: തിരുവനന്തപുരം നഗരത്തിൽ, തിരുവനന്തപുരം നഗരത്തിൽ. തിരുവനന്തപുരം നഗരത്തിൽ തിരുവനന്തപുരം നഗരത്തിൽ തിരുവനന്തപുരം നഗരത്തിൽ. തിരുവനന്തപുരം നഗരത്തിൽ തിരുവനന്തപുരം നഗരത്തിൽ തിരുവനന്തപുരം നഗരത്തിൽ. തിരുവനന്തപുരം നഗരത്തിൽ തിരുവനന്തപുരം നഗരത്തിൽ തിരുവനന്തപുരം നഗരത്തിൽ.

Single & Multi-Colour Screen Printing Operations

Practical stencil preparation, squeegee handling, registration techniques, and safety practices.

CO4 Target

Bloom: Apply

Screen Printing Lab

Not specified in the supplied official syllabus PDF — pending module/practical/reference pages: Practical experiment breakdown for Module IV is pending official syllabus job list. Content below is based on confirmed CO4 & Objective 6.

4.1 Screen Printing Frame, Mesh, Emulsion & Stencil Preparation–

Screen printing utilizes a wooden or aluminum frame stretched with synthetic polyester mesh. A photo-polymer emulsion is coated on the mesh, dried, exposed to UV light under a film positive, and washed with water to open the image stencil area.

Malayalam explanation: സ്ക്രീൻ പ്രിന്റിംഗ് ഫ്രെയിം, മെഷ്, എമുൽഷൻ, എക്സ്പോഷറും സ്റ്റൻസിൽ തയ്യാറാക്കുന്നതിനെക്കുറിച്ചാണ്. ഫ്രെയിം വുഡൻ അല്ലെങ്കിൽ അലൂമിനിയം ആയിരിക്കാം. മെഷ് സിന്ത്ഥെറ്റിക് പോളിസ്റ്റർ ആയിരിക്കും. എമുൽഷൻ ഫോട്ടോ-പോളിമർ ആയിരിക്കും. എക്സ്പോഷർ UV ലൈറ്റ് കീഴിൽ ഫിലിം പോസിറ്റീവ് ഉപയോഗിച്ച് നടത്തുന്നു. സ്റ്റൻസിൽ തയ്യാറാക്കുന്നത് വാഷിംഗ് വെള്ളം ഉപയോഗിച്ച് ചെയ്യുന്നു. Technical terms: *Frame, Mesh, Emulsion, Exposure, Stencil*.

4.2 Registration Controls & Multi-Colour Screen Printing–

Registration ensures exact alignment of images, especially during multi-colour printing. Mechanical stops or registration pins hold the substrate in the exact same position for each successive colour pass.

- 1 Prepare separate film positives and exposed screens for each colour.
- 2 Fix registration lay-guides on the printing bed.
- 3 Print the first colour, allow to dry completely.
- 4 Align the second screen with registration marks and print the second colour.

Malayalam explanation: മൾട്ടി-കോളർ സ്ക്രീൻ പ്രിന്റിംഗിൽ ചിത്രങ്ങൾ കൃത്യമായി അലൈൻ ചെയ്യുന്നതിനാണ് റജിസ്ട്രേഷൻ നിയന്ത്രണം. മെക്കാനിക്കൽ സ്റ്റോപ്പുകൾ അല്ലെങ്കിൽ റജിസ്ട്രേഷൻ പിൻസ് സബ്സ്ട്രേറ്റ് ഒരേ സ്ഥാനത്ത് തുടർച്ചയായി നിർമ്മിക്കുന്നു. Technical terms: *Registration, Alignment, Substrate*.

Standard Screen Printing Lab Operation Procedure

Not specified in the supplied official syllabus PDF — pending

module/practical/reference pages: The specific experiment job titles and detailed step lists will be updated upon receipt of the official SITTTTR practical manual. The standardized procedure below provides the essential operational foundation for CO4.

Standard Operating Procedure: Single-Colour Screen Printing

Aim: To prepare a screen stencil, set up registration guides, and perform single-colour screen printing on paper/cardboard substrate with proper safety practices.

Apparatus & Materials: Screen frame with polyester mesh, photo emulsion, sensitizer, UV exposure unit, water spray gun, squeegee (70° shore hardness), screen printing ink, substrate, apron, gloves.

1 Mesh Degreasing & Coating:

Clean
the
mesh
with
degreaser.
Apply
photo
emulsion
evenly
on
both
sides
using
a
scoop
coater
in
yellow/safe
light.

2 Drying & Exposure:

Dry
the
coated
screen
horizontally
in a
dark
drying
cabinet.
Place

the
film
positive
on
the
glass
bed
and
expose
to
UV
light
for
the
calculated
duration.

3 Development & Wash-out:

Spray
cold
water
gently
to
wash
away
unexposed
emulsion,
revealing
the
open
mesh
stencil.
Dry
the
screen
completely.

4 Screen Setup & Registration:

Clamp
the
screen
in
the
printing
bed.
Position
lay-
guides
on

the
bed
so
the
substrate
aligns
precisely
under
the
stencil.

5 Ink Application & Squeegee Pass:

Apply
ink
along
the
top
border
of
the
screen.
Hold
the
squeegee
at a
45°
to
60°
angle
and
draw
across
with
uniform
downward
pressure.

6 Drying & Reclaiming:

Place
printed
sheets
on
drying
racks.
Clean
remaining
ink
from
screen

using
solvent
and
wash
with
reclaiming
chemical.

Result: Uniform, sharp screen print obtained with crisp edges and proper ink coverage.

Diagrams, Flowcharts, Animations and Course Calculators

Prepress → Press → Postpress Workflow



Figure 1: The three major sequential stages of print manufacturing.

Screen Printing Setup & Component Identification

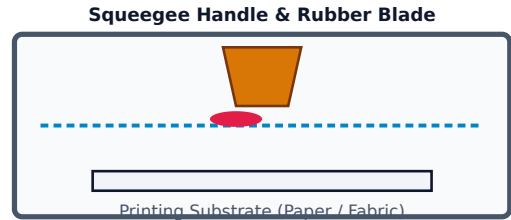
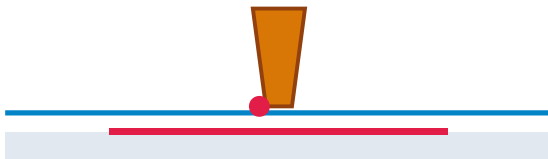


Figure 2: Component identification of screen frame, mesh, ink, squeegee, and substrate.

Interactive Animations

Screen Printing Ink Stroke Animation



Demonstrates squeegee action forcing ink through open mesh pores onto the substrate.

Prepress to Press Progression Animation



Visualizes digital design files transferring to mechanical press production.

Course Calculators

Self-Learning & Credit Calculator

Compute credits from scheme L:T:P:S hours.

Lecture Hours (L)

Tutorial Hours (T)

Practical Hours (P)

$$\text{Credits} = L + T + 0.5(P)$$

Screen Mesh Count Estimator

Recommend mesh count (threads/cm) by job type.

Job Substrate & Type

Select job type to see recommended mesh count.

Practical Step-by-Step Industry Walkthroughs

Walkthrough 1: Complete Prepress-to-Postpress Production Sequence

1 Graphic Design:

Create vector graphics and layout in publishing software.

2 Colour Separation:

Separate artwork into CMYK process color channels.

3 Plate/Stencil Prep:

Output film positives or image CTP plates.

4 Press Setup:

Mount plates/screens, meter ink, set registration guides.

5 Printing Pass:

Run press, monitor ink

density
and
registration.

6 Finishing:

Cut
to
final
size,
fold,
bind,
and
inspect
quality.

Walkthrough 2: Two-Colour Screen Printing Registration

1 Film Alignment:

Print
registration
crosshairs
on
artwork
corners.

2 First Screen Exposure:

Expose
Screen
1
(Base
Colour)
using
Positive
1.

3 Second Screen Exposure:

Expose
Screen
2
(Top
Colour)
using
Positive
2.

4 Bed Alignment:

Align
Screen

1 to
registration
marks
on
bed
lay-
stops.

5 First Pass:

Print
Colour
1
and
dry.

6 Second Pass:

Lock
Screen
2
into
lay-
stops,
match
crosshairs,
print
Colour
2.

PRACTICAL ACTIVITY CENTRE

Laboratory Discipline, Safety & Practical Exercises

Not specified in the supplied official syllabus PDF — pending module/practical/reference pages: The official SITTTTR experiment job list for 1101 (3 practical hours/week) is pending syllabus page release. Practical exercises will be updated upon receipt of official job lists.

Essential Laboratory Conduct Checklist

☒ Wear lab apron and safety leather footwear.

☒ Maintain clean work tables and degreased screens.

☒ Store solvents in labeled, fire-safe containers.

☒ Work collaboratively in assigned lab teams.

Teamwork & Laboratory Discipline (CO3/CO4)

Screen printing relies on disciplined teamwork: one student manages substrate feeding, one operates the squeegee, and one transfers printed sheets to drying racks while observing safety protocols.

Syllabus-Based Practice Questions with Answers

Question Bank Declaration: These expected practice questions are compiled strictly from confirmed Objectives 1-6 and Course Outcomes CO1-CO4. They are practice aids, not official SITTTTR exam papers.

Objective / 1-Mark Questions

1. Define Prepress.
2. What does CMYK stand for?
3. Name the tool used to push ink through a screen mesh.
4. State the primary function of Postpress.
5. What material is commonly used for modern screen mesh?
6. Name one application of screen printing in electronics.
7. What is registration in multi-colour printing?
8. State the Bloom level for CO4.

View Answers

1. Activities prior to printing (design, plate/stencil prep);
2. Cyan, Magenta, Yellow, Key (Black);
3. Squeegee;
4. Finishing & binding operations;
5. Synthetic polyester;
6. Printed Circuit Board (PCB) resist printing;
7. Precise alignment of colours/images;
8. Apply.

Short Answer (3 Marks) Questions

1. Explain the role of printing in packaging and logistics.
2. Differentiate between Relief and Planographic printing.
3. List three mandatory laboratory safety rules during screen printing.
4. Describe the function of photo emulsion in stencil making.

View Model Points

1. Protects goods, provides barcodes/shipping labels, displays brand info.
2. Relief has raised image areas; Planographic is flat using oil/water repulsion.
3. Wear apron/shoes, ensure solvent ventilation, keep long hair tied near machines.
4. Light-sensitive coating that hardens under UV, leaving unexposed stencil areas open for ink.

Detailed / 7-Marks Questions

1. Explain the complete printing workflow across Prepress, Press, and Postpress stages.
2. Describe the step-by-step procedure for preparing a screen printing stencil and performing single-colour printing.

View Model Points

1. Prepress: Graphic design, CMYK separation, imposition, plate/stencil prep. Press: Mounting carrier, ink application, impression cylinder pass onto substrate. Postpress: Cutting, folding, stitching, trimming, quality inspection.

2. Steps: Degrease mesh → Coat emulsion → Dry in dark → UV exposure with film positive → Water washout stencil → Clamp in bed → Align registration → Squeegee ink stroke → Dry print → Clean screen.

Sample Examination Paper — Based on Teaching & Assessment Scheme

Course 1101 — Introduction to Printing Technology

Time: 2½ Hours | Maximum: 60 Marks (ESE Theory Scheme)

Part A — Answer all questions (8 × 1 = 8 Marks)

- | | | |
|---|---|---|
| 1 | Define Prepress workflow. | 1 |
| 2 | Name four process colors in CMYK. | 1 |
| 3 | Identify the image carrier in Screen Printing. | 1 |
| 4 | Define Registration in printing. | 1 |
| 5 | State one function of a Squeegee. | 1 |
| 6 | Name one Postpress finishing operation. | 1 |
| 7 | State one safety precaution when handling lab solvents. | 1 |
| 8 | State the total contact hours for Course 1101. | 1 |

Part B — Answer short questions (8 × 3 = 24 Marks)

9. Explain the role of printing in daily life and education.
10. Describe the main activities involved in Prepress.
11. Compare Offset printing and Screen printing.
12. Explain the importance of laboratory discipline and safety.
13. Describe photo emulsion coating on screen mesh.
14. Explain how multi-colour registration is achieved in screen printing.

Part C — Answer detailed questions (4 × 7 = 28 Marks)

- 15A.** Describe the sequential printing workflow from Prepress to Postpress with a neat flowchart. **OR 15B.** Explain major printing processes classified by image carrier surface.
- 16A.** Describe step-by-step screen stencil preparation and squeegee printing operations. **OR 16B.** Detail laboratory safety rules, chemical handling, and emergency precautions for printing labs.

MASTER ANSWER KEY

Answer Direction for Practice Model Paper

1. Activities prior to printing (design, layout, plate/stencil prep).
2. Cyan, Magenta, Yellow, Black (Key).
3. Stencil on porous polyester mesh screen.
4. Precise alignment of images/colours on substrate.
5. Pushes and forces ink through open mesh pores.
6. Cutting / Folding / Binding / Lamination.
7. Ensure proper room ventilation and wear gloves/apron.
8. 90 Hours.

QUICK REVISION

Last-Day Summary Cards

Role of Printing

- Communication & media
- Packaging & labeling
- Advertising & logistics

Workflow Stages

- **Prepress:** Design & Plate
- **Press:** Ink Transfer
- **Postpress:** Cut & Bind

Printing Processes

- Relief (Flexo)
- Planographic (Offset)
- Intaglio (Gravure)
- Stencil (Screen)

Screen Lab Safety

- Apron & safety footwear
- Solvent ventilation
- Registration stops

BILINGUAL GLOSSARY

Printing Technology Technical Terms (English & Malayalam)

Printing Terminology Summary

English Technical Term	Malayalam Explanation (മലയാളം വിശദീകരണം)	Industry Definition
Prepress	പ്രിന്റിംഗ് പ്രക്രിയയുടെ തുടക്കത്തിൽ നിന്നും പ്രിന്റിംഗ് പ്രസ്സിലേക്ക് (പ്രിന്റിംഗ്, പ്രസ്സ്) പ്രവേശിക്കുന്നവരെ ഉൾക്കൊള്ളുന്നു.	All preparation steps performed prior to printing on press.
Press	പ്രിന്റിംഗ് പ്രക്രിയയുടെ തുടക്കത്തിൽ നിന്നും പ്രിന്റിംഗ് പ്രസ്സിലേക്ക് (പ്രിന്റിംഗ്, പ്രസ്സ്) പ്രവേശിക്കുന്നവരെ ഉൾക്കൊള്ളുന്നു.	The actual printing phase where ink is transferred to a substrate.
Postpress	പ്രിന്റിംഗ് പ്രക്രിയയുടെ അവസാനത്തിൽ നിന്നും പ്രിന്റിംഗ് പ്രസ്സിലേക്ക് (പ്രിന്റിംഗ്, പ്രസ്സ്) പ്രവേശിക്കുന്നവരെ ഉൾക്കൊള്ളുന്നു.	Finishing operations such as cutting, folding, binding, and trimming.
Screen Printing	പ്രിന്റിംഗ് പ്രക്രിയയുടെ തുടക്കത്തിൽ നിന്നും പ്രിന്റിംഗ് പ്രസ്സിലേക്ക് (പ്രിന്റിംഗ്, പ്രസ്സ്) പ്രവേശിക്കുന്നവരെ ഉൾക്കൊള്ളുന്നു.	Stencil printing process using a squeegee to force ink through mesh.
Squeegee	പ്രിന്റിംഗ് പ്രക്രിയയുടെ തുടക്കത്തിൽ നിന്നും പ്രിന്റിംഗ് പ്രസ്സിലേക്ക് (പ്രിന്റിംഗ്, പ്രസ്സ്) പ്രവേശിക്കുന്നവരെ ഉൾക്കൊള്ളുന്നു.	Rubber or polyurethane blade used to push ink across the screen.
Mesh	പ്രിന്റിംഗ് പ്രക്രിയയുടെ തുടക്കത്തിൽ നിന്നും പ്രിന്റിംഗ് പ്രസ്സിലേക്ക് (പ്രിന്റിംഗ്, പ്രസ്സ്) പ്രവേശിക്കുന്നവരെ ഉൾക്കൊള്ളുന്നു.	Woven synthetic fabric stretched tightly across a screen frame.
Registration	പ്രിന്റിംഗ് പ്രക്രിയയുടെ തുടക്കത്തിൽ നിന്നും പ്രിന്റിംഗ് പ്രസ്സിലേക്ക് (പ്രിന്റിംഗ്, പ്രസ്സ്) പ്രവേശിക്കുന്നവരെ ഉൾക്കൊള്ളുന്നു.	Precise positioning of print images during multi-colour printing.
Emulsion	പ്രിന്റിംഗ് പ്രക്രിയയുടെ തുടക്കത്തിൽ നിന്നും പ്രിന്റിംഗ് പ്രസ്സിലേക്ക് (പ്രിന്റിംഗ്, പ്രസ്സ്) പ്രവേശിക്കുന്നവരെ ഉൾക്കൊള്ളുന്നു.	Light-sensitive chemical coating used to create screen stencils.
CMYK	പ്രിന്റിംഗ് പ്രക്രിയയുടെ തുടക്കത്തിൽ നിന്നും പ്രിന്റിംഗ് പ്രസ്സിലേക്ക് (പ്രിന്റിംഗ്, പ്രസ്സ്) പ്രവേശിക്കുന്നവരെ ഉൾക്കൊള്ളുന്നു.	Cyan, Magenta, Yellow, and Key (Black) process colors.
Substrate	പ്രിന്റിംഗ് പ്രക്രിയയുടെ തുടക്കത്തിൽ നിന്നും പ്രിന്റിംഗ് പ്രസ്സിലേക്ക് (പ്രിന്റിംഗ്, പ്രസ്സ്) പ്രവേശിക്കുന്നവരെ ഉൾക്കൊള്ളുന്നു.	Any material (paper, cardboard, fabric, plastic) being printed upon.

LEARNING RESOURCES & REFERENCES

Official Syllabus Reference Declaration

Not specified in the supplied official syllabus PDF — pending reference pages:

Prescribed textbooks, digital learning resources, and official reference books for Course 1101 are pending syllabus page release.

Official Scope Declaration: Academic scope is strictly governed by the official SITTR Kerala Diploma Revision 2026 Course Description for Subject Code 1101.